PERSONALIZED AI FOOD EXPIRY PREDICTOR

Project Id: 25-26J-351

Project Proposal Report

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BSc (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

....27/08/2025

Signature of the supervisor Date

ABSTRACT

Commonly worldwide growing problem is food waste, it is caused by forgetting the expiry dates of items or inaccurate generic shelf-life estimates. Food is stored in different ways by many people and consumed in different speeds, which makes it difficult to know exact date of something will go bad. Most of the existing expiry tracking apps use fixed dates or complex systems which makes the users hard to adopt. Main goal of this project is to develop an AI-powered food expiry predictor that adapts to each user's unique habits. Simple machine learning model will be used in this system to learn user's previous consumption and spoilage patterns. It considers details such as the type of food, storage method, and average usage behavior. Using these details, it predicts personalized expiry dates instead of depending only on generic shelf-life data. The application allows users to log food items, receive reminders before expiry, and mark items as consumed or spoiled. Every time the user provides feedback, the system updates its predictions, becoming more accurate over time. The approach is lightweight, easy to use, and requires no complex image processing or expensive sensors, making it accessible to everyone. By giving frequent and accurate reminders, this system helps reduce food waste, saves money, and improves household food management while being sustainable. Businesses can also use such a system to analyze consumption patterns and adjust stocking strategies. To put it briefly, this project builds a smart, user-friendly assistant that learns from individual behavior to predict food expiry dates more accurately, ultimately making food management easier and reducing waste.

Keywords: *Food waste, Expiry prediction, Machine learning, Generic shelf-life, Consumption patterns, Storage methods, Personalized system, Sustainability, Expiry reminders*

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LIST OF ABBREVIATIONS

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **NLP** – Natural Language Processing
- **IoT** – Internet of Things
- **FAO** – Food and Agriculture Organization
- **API** – Application Programming Interface

1. INTRODUCTION

The AI-Powered Kitchen Ecosystem is a comprehensive research initiative designed to address the global challenges of food waste and poor nutrition through an integrated, AI-driven platform. This holistic system seamlessly combines four specialized modules: a Spontaneous Cooking Assistant that uses advanced computer vision to identify available ingredients and suggest relevant recipes, an AI Shopping Agent that provides voice-enabled, intelligent product search and cross-platform price comparisons, a Behavioral Food Expiry Predictor that learns from user habits to deliver personalized spoilage alerts, and a Nutritional Guidance and Predictive Health Intervention System that provides AI-powered, personalized dietary and wellness recommendations. Together, these components create a synergistic workflow that supports the entire food lifecycle, from planning and purchasing to cooking and consumption, fostering sustainable household practices and reducing waste [22]–[24].

Within this ecosystem, the Food Expiry Predictor plays a pivotal role by focusing on personalized food spoilage estimation. Unlike generic “use by” or “best before” labels, this component leverages lightweight AI models to provide expiry predictions that reflect actual household usage and storage conditions. It operates using simple inputs such as purchase dates, food types, and storage methods, while progressively adapting to user-specific consumption patterns. By acting as an intelligent assistant within the broader system, the Food Expiry Predictor enhances household food management by giving users timely and personalized insights into freshness, thereby improving safety, promoting sustainability, and reducing food waste [25]–[27].

1.1 Background & Literature Survey

Household food waste is a serious global issue: according to the FAO, about one-third of all food produced for human consumption approximately 1.3 billion tonnes have lost or wasted each year [1], [2]. This waste stems from factors such as forgotten items, improper storage, or reliance on generic “use by” labels that do not reflect how people actually use their food.

Traditional expiry tracking method such as printed dates and online shelf-life table are inherently static and unable to adapt to variations in household behavior, storage conditions, or environmental factors [3], [4]. While mobile apps and manual tracking tools offer some help, they typically employ generic expiration data and lack the ability to tailor predictions to individual usage patterns, often leading to premature disposal of still-edible food or unanticipated spoilage.

Machine learning offers a solution by enabling personalized expiry predictions based on user-specific data such as purchase dates, storage locations, and consumption history which can more accurately reflect actual food longevity in each household. This is particularly valuable where environmental factors contribute significantly to deviation from standard shelf-life expectations [5].

The proposed system aims to fill this gap by developing an AI-powered food expiry predictor that adapts to each user's habits and conditions over time. Unlike complex IoT-based solutions, this approach uses lightweight, accessible ML models that operate using manual user inputs and do not require specialized hardware making it both affordable and practical for everyday households while significantly reducing avoidable food waste.

Effective management of household food storage, particularly in predicting and preventing food spoilage, is critical to reducing global food waste and ensuring food safety. The emergence of artificial intelligence and machine learning presents new opportunities for improving the accuracy and relevance of food expiry predictions. These technologies enable the creation of systems that adapt to user-specific habits and environmental conditions, thereby offering more practical and personalized solutions compared to generic expiry labeling methods.

Food waste remains a global challenge, with approximately 931 million tonnes wasted annually, according to the United Nations Environment Programme [6]. While traditional methods such as printed expiry dates and mobile reminder applications are widely used, they often fail to account for variations in storage conditions, product freshness at the time of purchase, and user consumption patterns [7]. As a result, these methods can be inaccurate,

leading to either premature disposal of edible food or delayed recognition of spoilage, which poses health risks [8].

Recent studies have explored IoT-based food monitoring systems, integrating sensors to measure temperature, humidity, and other environmental parameters [9]. While these systems provide accurate real-time monitoring, they are often expensive, require specialized hardware, and are impractical for everyday households [10]. Similarly, some ML-based approaches have been proposed to predict expiry dates; however, many rely on generic datasets and do not adapt to individual user conditions, limiting their effectiveness [11].

The proposed personalized food expiry predictor aims to address this gap by leveraging lightweight machine learning models trained on user-specific data such as purchase dates, storage locations, and historical consumption trends. Unlike complex IoT-based solutions, this approach eliminates the need for additional hardware, making it cost-effective and accessible to a broader audience [12]. By learning and adapting over time, the system can offer tailored expiry predictions that reflect real household conditions, reducing avoidable food waste and improving food safety.

Integrating personalization into AI-driven expiry prediction systems offers a significant advancement over traditional approaches. It not only enhances prediction accuracy but also promotes sustainable consumption practices by empowering users with reliable, adaptive, and context-aware insights

1.2 Research Gap

Most existing expiry tracking systems rely solely on static labels or generic shelf-life tables, which do not account for individual household practices, storage conditions, or environmental factors often resulting in premature disposal or unexpected spoilage [13], [14]. Some mobile applications incorporate barcode or receipt logging but still depend on generic data, lacking per-user adaptation and learning over time [15]. Techniques using computer vision to estimate freshness like detecting ripeness or defects via image features improve real-time assessment but remain heavy in model complexity and still fail to personalize predictions based on user behavior [16]. IoT-based or smart-fridge systems offer detailed monitoring through sensors, but their high cost and dependency on specialized hardware make them impractical for widespread home adoption [17]. Furthermore, despite growing interest in ML-driven shelf-life prediction, most models operate with global datasets and lack incremental learning or per-user customization needed for home environments [18]. Therefore, there is a clear research and practical gap: a cost-effective, personalized AI expiry predictor is missing one that learns from minimal user input such as food type, purchase date, storage location, provides transparent expiry estimates, adapts dynamically to each user's patterns, and functions without expensive sensors or infrastructure.

1.2.1 Comparison with Existing Systems

System / Approach	Features	Limitations	Novelty of Proposed System
Printed Expiry Dates & Shelf-Life Tables [28]	Static "use by" or "best before" labels provided by manufacturers.	Cannot adapt to household-specific storage/usage conditions. Often leads to premature disposal or unexpected spoilage.	Learns from each user's consumption and storage patterns to predict adaptive expiry dates.
Generic Mobile Apps (e.g., FoodKeeper App – USDA) [29]	Users manually log items and get reminders based on generic shelf-life databases.	Relies on fixed generic data. No personalization.	Provides personalized expiry predictions based on individual user behavior and habits.

Barcode/Receipt Tracking Apps [30]	Track purchases via scanning, link to generic databases for shelf-life.	Still generic, does not learn from user-specific consumption speed.	Adaptive ML models improve accuracy by learning from feedback and spoilage reports.
Computer Vision Approaches (Freshness Detection) [31]	Uses image processing to estimate ripeness or detect spoilage.	Heavy models, requires photos, complex, not user-friendly, and does not adapt over time.	Lightweight ML system requiring only simple user inputs (purchase date, storage type, usage).
IoT/Smart Fridge Systems [32]	Use sensors (temperature, humidity, etc.) to monitor storage conditions.	Expensive, hardware-dependent, limited adoption in average households.	No IoT devices needed, cost-effective, accessible for everyday households.
Proposed Personalized AI Expiry Predictor	Predicts expiry using ML on user-specific consumption/storage data. Provides reminders, reports, and adapts continuously.	–	Combines personalization, adaptive learning, low cost, and high accessibility—unlike any existing system.

Table 1.1 --- Comparison Table

1.3 Research Problem

The main challenge lies in developing an AI system capable of accurately predicting food expiry dates based on individual user behavior rather than relying solely on generic shelf-life data. Current solutions do not adapt to household-specific factors such as storage habits, consumption speed, and environmental conditions, resulting in inaccurate predictions and unnecessary food waste. The objective of this research is to develop a personalized AI food expiry predictor that learns from user-specific patterns, such as how quickly different food items are consumed, and incorporates contextual factors like storage location and local climate to improve prediction accuracy. The system aims to provide adaptive expiry estimates that update over time as more user data is collected, while using lightweight machine learning models to ensure it remains practical and accessible for everyday users

without the need for specialized hardware or complex IoT devices. By addressing these challenges, the system aims to reduce avoidable food waste, improve household food management, and demonstrate that personalized, user-centric AI models can outperform generic expiry trackers in real-world settings.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this project is to design and implement a personalized AI-based food expiry prediction system that enables users to efficiently track and manage their groceries while reducing food waste. The system will allow users to input essential details of food items, including the name, purchase date, and storage location such as fridge, freezer, or pantry. By integrating historical shelf-life datasets with user-specific consumption patterns, the system will generate accurate, adaptive expiry predictions that improve over time through continuous learning. A lightweight machine learning model will be developed to ensure the solution remains efficient, accessible, and operational without specialized hardware or complex infrastructure. The system will incorporate adaptive mechanisms to account for environmental factors such as local climate and individual storage conditions, ensuring greater prediction reliability. Furthermore, an intuitive and user-friendly interface will be designed to allow users to easily add, view, and manage their food inventory, supported by real-time notifications and alerts to prevent spoilage. Through these objectives, the project aims to provide a cost-effective, intelligent, and practical solution that directly addresses the gap in current expiry tracking technologies by combining personalization, adaptability, and ease of use.

2.2 Specific Objectives

- To design a user-friendly interface that allows users to input and track food details such as name, purchase date, storage location, and quantity.

- To build and integrate a lightweight machine learning model that predicts expiry dates based on food type, storage conditions, and user-specific consumption patterns.

• To implement a cloud-based database for storing user inputs, historical data, and adaptive learning parameters.

• To enable personalized learning by continuously updating expiry predictions as the system learns from each user's storage and consumption behavior.

• To develop a backend system for handling ML predictions, API requests, and data synchronization between frontend and database.

• To implement a notification and reminder system (that alerts users about upcoming expiries through scheduled reminders.

• To ensure the solution is lightweight, scalable, and cost-effective, avoiding reliance on external IoT devices or heavy AI frameworks.

• To evaluate the system's performance in terms of accuracy, usability, and impact on food waste reduction, ensuring practical benefits for households.

3. METHODOLOGY

3.1 Overall System Architecture

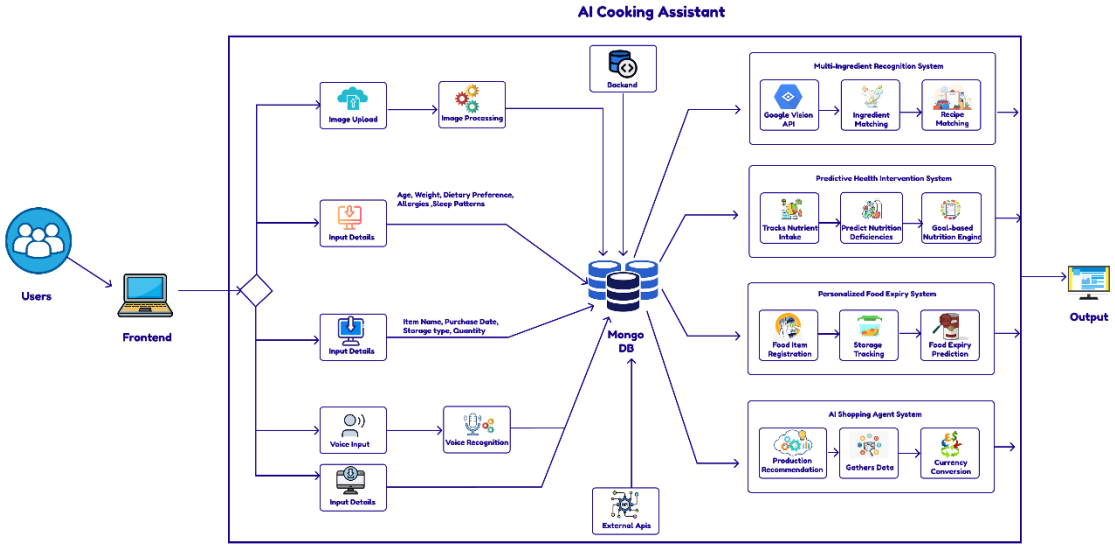

Figure 3.1 – Overall System Architecture

3.1.1 Food Expiry System Architecture

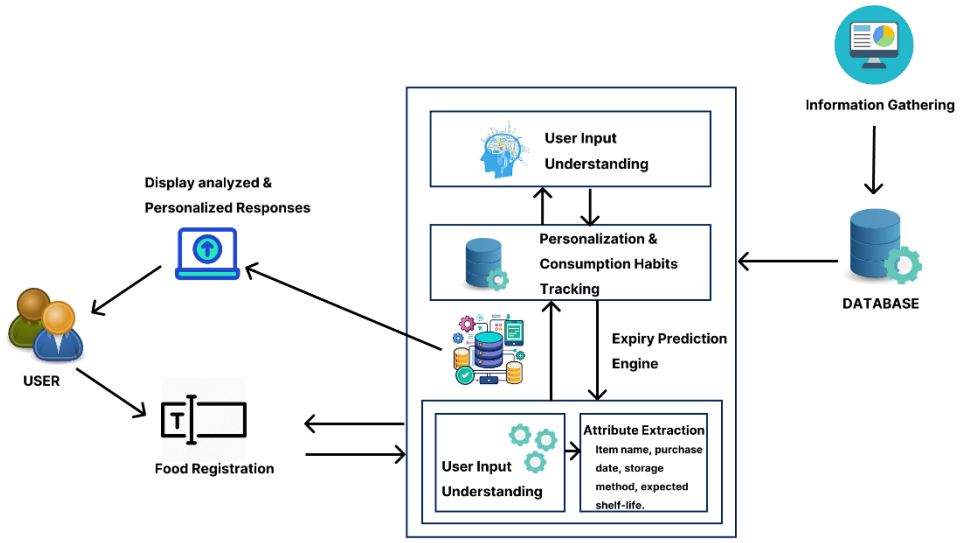

Figure 3.2 – Food Expiry System Architecture

3.2 Tools and Technologies

- Python – For machine learning and data processing
- scikit-learn – For building simple ML models (Linear Regression, Decision Tree)
- pandas / NumPy – For handling and analyzing data
- Flask – Backend API to serve ML predictions
- Node.js + Express – Backend server and API management
- React.js – Frontend interface for user interaction
- MongoDB Atlas – Cloud-based NoSQL database
- joblib / pickle – For saving and loading ML models
- node-cron – For scheduling expiry reminders
- SendGrid / Nodemailer – For sending email notifications
- Google Colab / Jupyter Notebook – ML prototyping and testing

3.3 Commercialization

The proposed AI-powered personalized food expiry predictor is a lightweight and affordable application that uses machine learning to generate accurate expiry predictions based on user-specific purchase dates, storage conditions, and consumption patterns, reducing reliance on generic shelf-life tables [19]. With strong commercialization potential in both household (B2C) and small business (B2B) markets, the system can be monetized through a freemium model, premium analytics, and SaaS licensing for inventory management. Its key benefits include cost savings by preventing premature food disposal, sustainability through reduced waste, and ease of adoption without requiring expensive IoT devices [20]. Scalable, adaptable, and capable of integration with grocery retailers and delivery platforms, this solution offers a financially viable and socially impactful opportunity for entrepreneurship while addressing a critical global issue [21].

3.3.1 Future Scope

- Recipe Integration: Extend the system to suggest recipes using near-expiry items, helping users consume food before it spoils.
- Barcode and Image Recognition: Allow users to scan products or take photos for automatic identification and expiry prediction.
- Smart Home Integration: Connect predictions with smart fridges or kitchen devices for automated tracking of food items.
- Population-Level Insights: Aggregate anonymized data to forecast regional food demand, waste trends, or supply chain needs.
- Personalized Nutrition Advice: Combine expiry predictions with dietary recommendations to suggest optimal meal plans based on user preferences and available food.

3.4 Budget and Budget Justification

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application.
Following is the budget justification for the whole research system.

Description	Cost (LKR)
Electricity	5000.00
Internet	5000.00
Communication	2000.00
Transportation	3000.00
Paper Publishing	15000.00
Backend Hosting	12000.00
Software Purchasing	10000.00
Prototype Materials (Basic Devices)	8000.00
Total	**60000.00**

Table 3.1 -- Budget Table

3.4.1 Budget Justification

- **Electricity (LKR 5,000):** Covers power consumption during system development, AI model training, and testing.
- **Internet (LKR 5,000):** High-speed internet for research, cloud services, and API access.
- **Communication (LKR 2,000):** Costs for phone calls, mobile data, and collaboration platform subscriptions to ensure seamless coordination among the research team
- **Transportation (LKR 3,000):** Travel for research meetings, presentations, or data collection if needed.
- **Paper Publishing (LKR 15,000):** Costs for conference or journal publication of the research findings.
- **Backend Hosting (LKR 12,000):** Cloud services for hosting the system and ensuring scalable access.
- **Software Purchasing (LKR 10,000):** Licenses for necessary software, IDEs, databases, and AI libraries not available for free.
- **Prototype Materials (LKR 8,000):** Funds for basic hardware components (e.g., sensors, microcontrollers) to build and test a physical proof-of-concept.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- User registration and authentication.

- Add food items with details (name, purchase date, storage location, quantity, expiry date if available).

- Edit, update, or delete food items.

- Predict expiry dates using ML models based on food type, purchase date, and storage method.

- Adapt expiry predictions over time based on user-specific storage and consumption habits.

- Store user data, food history, and prediction results in MongoDB Atlas.

- Send expiry reminders/alerts through email or app notifications.

- Allow customization of notification timings.

- Provide reports on food usage, expired items, and food waste reduction trends.

- Support Flask API for ML model predictions.

- Support Node.js + Express API for frontend-backend communication.

4.2 Non-functional requirements

- Predictions and responses should be generated quickly with faster performance.

- Must support increasing number of users and large datasets.

- Simple, intuitive, mobile-responsive interface for non-technical users for increased usability.

- System uptime of at least 99% with accurate and timely notifications.

- Encrypt all user data (in transit and at rest) and use secure authentication (JWT/OAuth).

• Modular, clean codebase with easily upgradable ML models.

• Deployable on cloud and compatible across devices/browsers.

• Use lightweight ML models that avoid need for IoT devices or expensive hardware which will be cost effective.

5. GANTT CHART

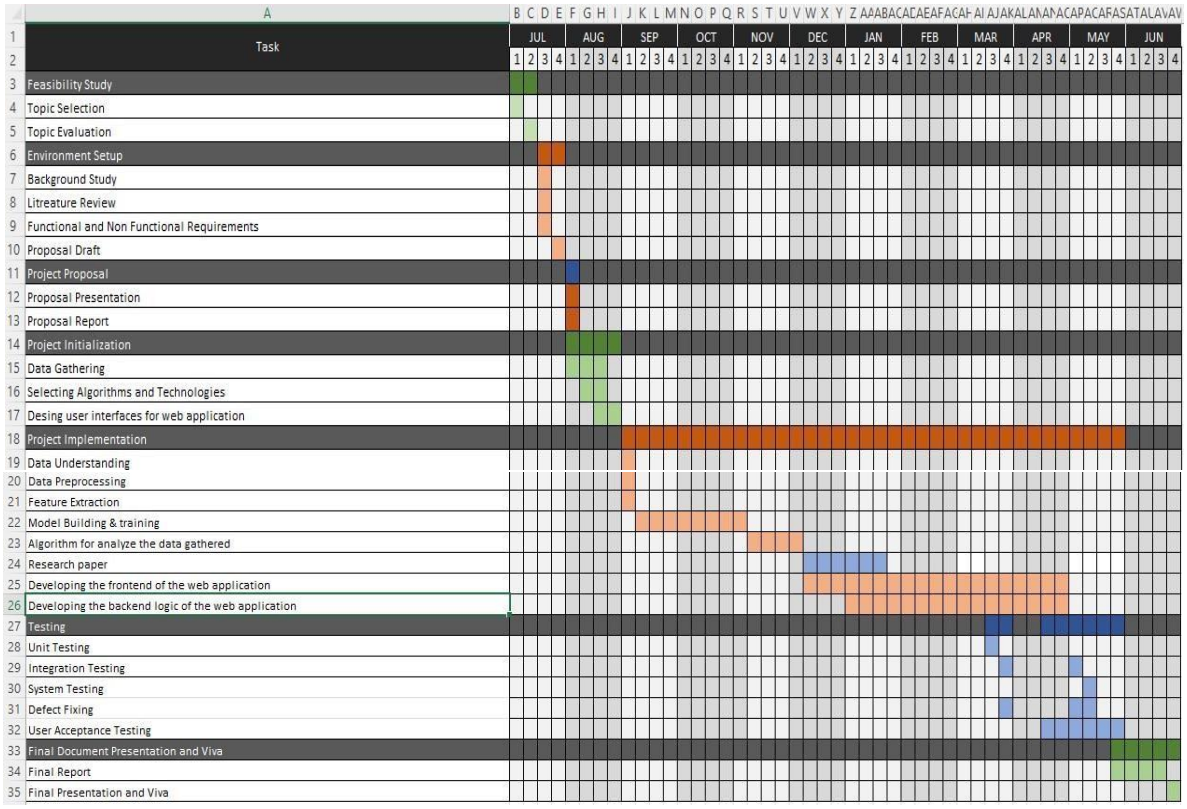

Figure 5.1 -- Gannt Chart

6. WORK BREAKDOWN CHART

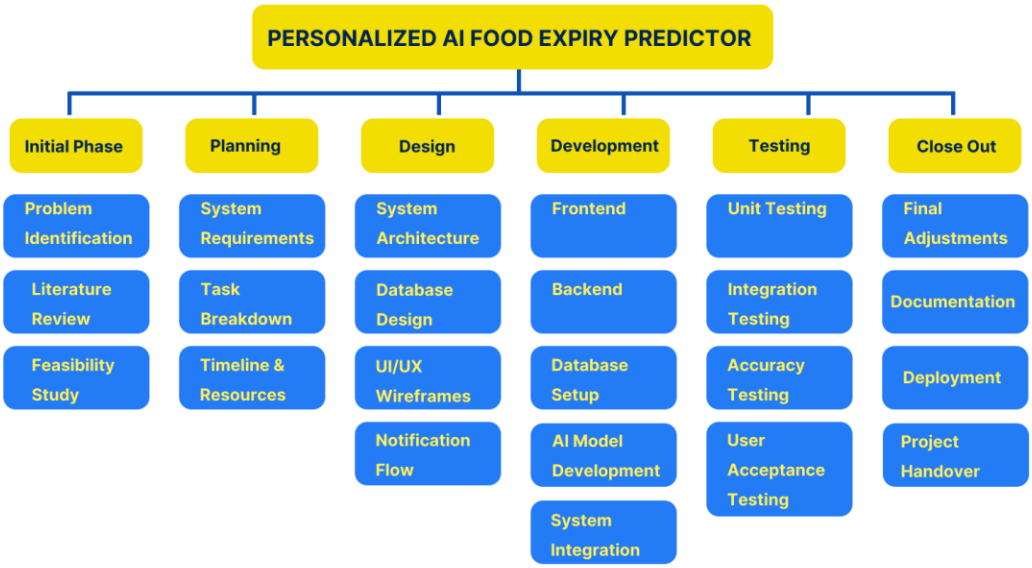

Figure 6.1 – Work Breakdown Chart

1. Initial Phase

This stage focuses on understanding the problem, reviewing prior work, and validating feasibility.

- Problem Identification: Define the core challenge of food waste due to poor expiry tracking and lack of personalized prediction systems.
- Literature Review: Analyze existing systems (expiry labels, mobile apps, IoT systems) and highlight their limitations to justify the proposed solution.
- Feasibility Study: Assess technical, financial, and operational feasibility to ensure the system can be realistically developed and adopted.

2. Planning

This stage involves systematic preparation for project execution.

- System Requirements: Collect both functional (expiry prediction, notifications, reminders) and non-functional (scalability, usability, accuracy) requirements.
- Task Breakdown: Decompose the project into manageable tasks (frontend, backend, AI model, testing, deployment).
- Timeline & Resources: Allocate time, budget, and human resources (developers, testers, domain experts) to ensure efficient project execution.

3. Design

Focuses on the blueprint of the system.

- System Architecture: Define the overall structure including client-side (React frontend), server-side (Node.js/Express backend), database (MongoDB/MySQL), and AI model integration.
- Database Design: Develop entity-relationship diagrams (ERD) to manage ingredients, expiry data, user profiles, and prediction history.
- UI/UX Wireframes: Create interactive prototypes for the mobile/web application, ensuring simplicity and user-friendliness.
- Notification Flow: Design the workflow for personalized expiry alerts, reminders, and health-related suggestions.

4. Development

This is the implementation phase where system modules are built.

- Frontend: Develop user interfaces (recipe suggestions, expiry tracking dashboard, notifications).
- Backend: Build APIs for data handling, authentication, and integration with the AI model.

- Database Setup: Implement the designed schema and configure databases for efficient storage and retrieval.
- AI Model Development: Train and integrate a machine learning model that predicts expiry dates based on user habits and storage patterns.
- System Integration: Ensure smooth interaction between frontend, backend, database, and AI modules.

5. Testing

Validates system functionality, performance, and accuracy.

- Unit Testing: Test individual components (UI forms, API endpoints, AI prediction module).
- Integration Testing: Validate the communication between system modules (frontend ↔ backend ↔ AI ↔ database).
- Accuracy Testing: Measure AI model performance using datasets (precision, recall, F1-score for expiry prediction).
- User Acceptance Testing (UAT): Gather feedback from real users to check usability, accuracy of predictions, and notification effectiveness.

6. Close Out

Final deployment and project closure activities.

- Final Adjustments: Fix remaining bugs, optimize performance, and refine UI.
- Documentation: Prepare detailed technical documentation (system design, APIs, model details) and user manuals.
- Deployment: Release the system to production (web or mobile platforms).
- Project Handover: Transfer ownership to stakeholders, including training and knowledge transfer sessions.

7. DESCRIPTION OF PERSONNEL AND FACILITIES

7.1 Supervisor

Mrs. Lokesha Weerasinghe

Senior Lecturer

Faculty of Computing | Information Technology

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7.2 Co-Supervisor

Mrs. Chathurya Kumarapperuma

Assistant Lecturer

Faculty of Computing | Information Technology

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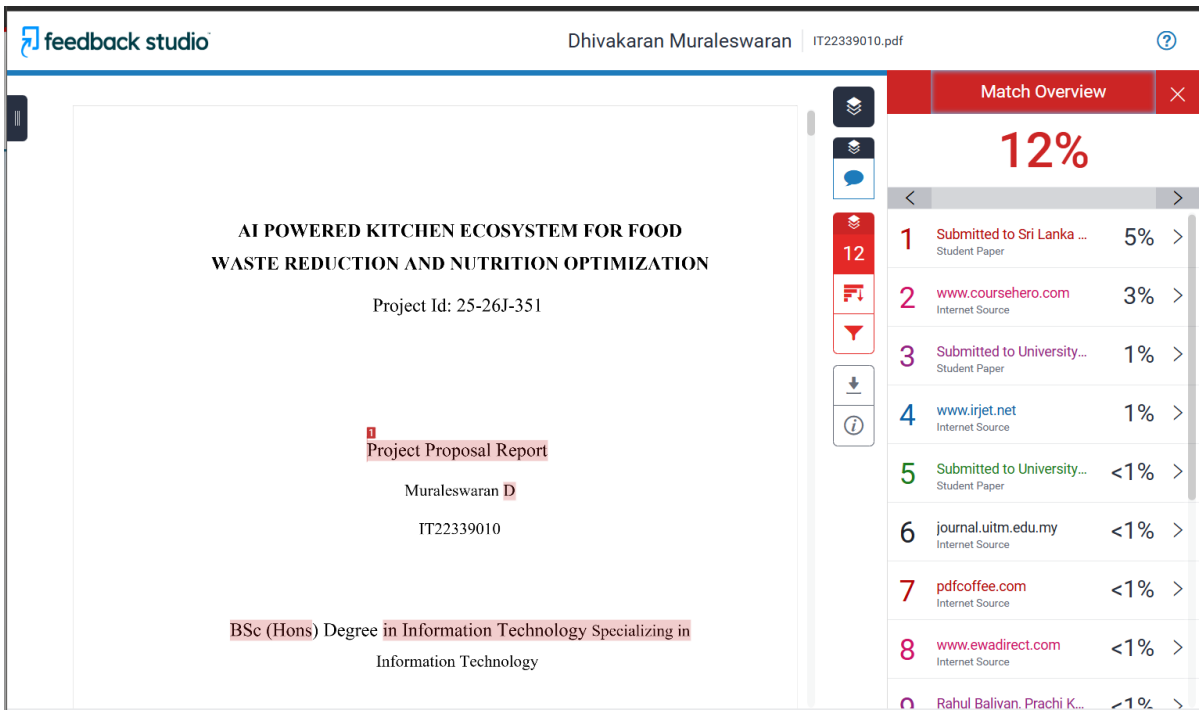
9. APPENDICES

Appendix - A: Plagiarism Report



Appendix 1 -- Plagiarism Report